	GLOBAL ECONOMICS **Grade: 10 - Learning Evidence 3** **DECISION-MAKING UNDER UNCERTAINTY** **Version 2C** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Learning objective: Compare decision criteria and their limitations, and construct and interpret decision trees for staged filtering under uncertainty.	**Assessment criteria:** **C8**: Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations. **C9**: Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative. **C10**: Evaluates economic and financial decision-making scenarios by selecting and justifying the most appropriate decision criterion according to the information available, risk attitude, and decision context.

Name: _____ **Grade:** _____ **Date:** _____

Criteria assessment

This exam assesses C8 and C9, each worth 6 points. A **minimum of 4 points per criterion** is required to pass. Show all relevant calculations and justify your answers. Partial credit may be awarded for correct procedures despite arithmetic errors; answers without demonstrated procedures receive no credit. A 1-point penalty per criterion applies if the student's name is missing or the assessment is submitted late. C10 is assessed separately.

Question	C8	C9
1	3	–
2	3	–
3	–	2
4	–	2
5	–	2
Total	**6**	**6**

1. Camera Rental Plans [C8-3]

A camera rental studio selects its inventory under uncertain demand. Profits are in thousand USD.

Alternative	High	Normal	Low
$P(S_j)$	0.2	0.5	0.3
Essential (A_1)	115	100	75
Shared (A_2)	155	165	45
Extensive (A_3)	265	55	-30
Modular (A_4)	190	145	-10

Decision statistics (Regret = maximum regret):

Alt.	Max	Min	Regret	EMV
A_1	115	75	150	95.5
A_2	165	45	110	127
A_3	265	-30	110	71.5
A_4	190	-10	85	107.5

(a) Using the original camera rental payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]**

(b) A subsidy s (integer, $0 \le s \le 100$) increases the low-demand payoffs of A_3 and A_4 by s. An alternative is eligible only if all its payoffs are nonnegative. Determine each camera rental plan's eligibility based on the value of s. **[0.5]**

(c) Among eligible camera rental plans, find the smallest allowed subsidy s at which the Maximin recommendation changes from A_1 to A_4 alone. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $0 \le s \le 100$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

2. Outdoor Cinema Refreshments [C8-3]

A cinema manager selects a refreshment kiosk under uncertain audience size. Profits are in hundred USD.

Alternative	High	Low
$P(S_j)$	0.3	0.7
Portable (C_1)	108	72
Full-service (C_2)	174	22
Deluxe (C_3)	204	-8
Fixed (C_4)	158	54

Decision statistics (Regret = maximum regret):

Alternative	Max	Min	Regret	EMV
C_1	108	72	96	82.8
C_2	174	22	50	67.6
C_3	204	-8	80	55.6
C_4	158	54	46	85.2

(a) Using the original refreshment kiosk payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]**

(b) A sponsor replaces C_1's large-audience payoff of 108 with g (integer, $108 \leq g \leq 180$). An alternative is eligible only if all its payoffs are at least 46. Determine each refreshment kiosk's eligibility based on the value of g. **[0.5]**

(c) Among eligible refreshment kiosks, find the smallest allowed sponsored payoff g at which the Minimax Regret recommendation changes from C_4 to C_1 alone. Recalculate regret benchmarks using only eligible kiosks. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $108 \leq g \leq 180$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

3. Garden Centre Stock [C9-2]

A garden centre selects a seedling stock plan D_1–D_6. Net profits are in hundred USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
D_1	164	68	72	137.6
D_2	188	44	48	154.4
D_3	140	92	96	123.2
D_4	212	20	72	154.4
D_5	116	68	120	99.2
D_6	236	-4	96	154.4

Apply these stages in order:

1. EMV/Bayes: Best 3.
2. Maximax: Best 2.
3. Maximin: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

4. Climbing Gym Expansion [C9-2]

A climbing gym selects an expansion plan F_1–F_6. Net profits are in thousand USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
F_1	213	59	55	154.7
F_2	169	103	99	144.8
F_3	191	81	77	164.6
F_4	268	4	99	138.2
F_5	147	92	121	125
F_6	235	37	66	145.9

Apply these stages in order:

1. Minimax Regret: Best 5.
2. EMV/Bayes: Best 4.
3. Maximax: Best 2.
4. Maximin: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

5. Art Materials Orders [C9-2]

An art materials distributor selects an order plan H_1–H_6. Net profits are in hundred USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
H_1	129	81	72	111.4
H_2	145	65	56	124.2
H_3	177	33	48	112.2
H_4	113	73	88	97
H_5	201	9	72	106.6
H_6	161	49	40	118.6

Apply these stages in order:

1. Maximin: Best 4.
2. Minimax Regret: Best 3.
3. Maximax: Best 2.
4. EMV/Bayes: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

C10: Separate activity reminder (unscored)

Complete the individual **Interactive Decision Webpage: C10 Practice** activity and submit the deployed webpage URL. See the separate activity for instructions.